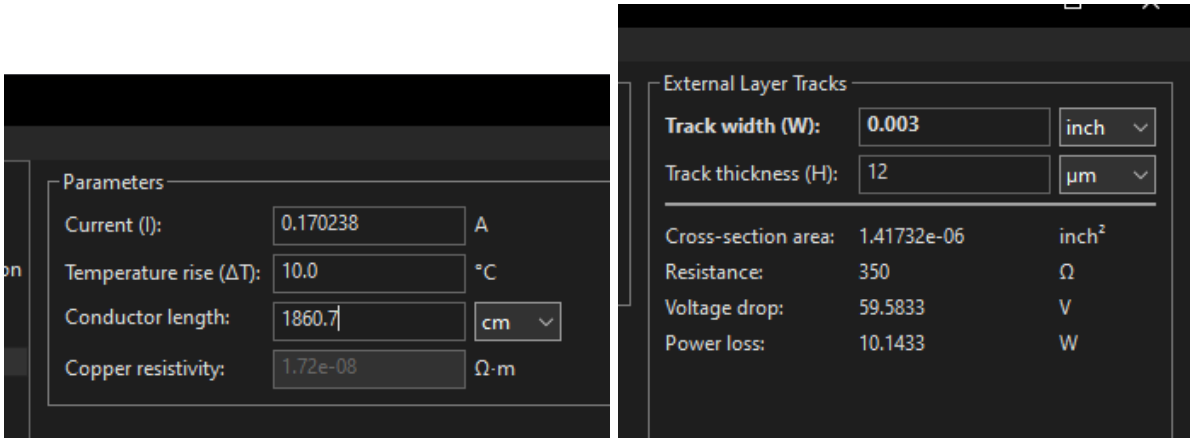
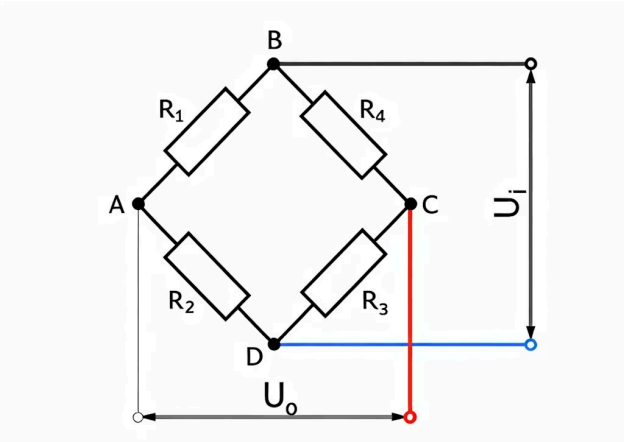


Here from the Kicad Calculator, You can see the required length of the Conductor to be **1,860.7cm** of Copper to reach 350 ohms of resistance, under the JLC minimum measurement constraints of **12µm (1/3 oz 3/3 mil)** (***safe*** minimum) that the both the **Track width (0.003inch / 3)** and **Track thickness (12µm).**



Other teams have had success using a full bridge style of gauge- stating that “Full-bridge configuration, where all resistors are active: This configuration compensates for the influence of bending and temperature changes, resulting in a voltage ratio Ui/U0 that is linear and dependent solely on the deformation caused by axial force.” Source

Deciding the Layout : under the JLC minimum measurement constraints of **12µm (1/3 oz 3/3 mil)** (***safe*** minimum) **Track width (3/ 3mil)** and **Track thickness (12µm).**

My Plan for the Gauge:

My gauge will by inspired by the gauge used by the FSAE team at University of Ljubljana, which utilized pairs of *HBM 1-XY11-3/350 Rosette Strain gauge for ferritic steel* To create a full bridge style gauge, with two of the T rosette - style gauges positioned on the suspension members to complete the bridge.

• It has two independent measuring grids (resistors) printed onto one single adhesive backing.

• The two grids are arranged perpendicular to each other in a "T" shape. One grid is aligned parallel to the tube to measure the primary axial stretching/compression, and the other is aligned perpendicular to capture lateral grid movement (Poisson's ratio).

Before Tackling T-style Rossette, I started with a single resistor design:



-this requires a rectangle that is 476.85mm x 6.62mm, which is well under JLC's minimums

Creation in Kicad:
After creating the Serpentine in Kicad, I left extra space in order to develop the future second resistor, so it ended up at roughly **483.5mm x 13.9mm**